# Harshavardhan P Kamarthi

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**Research Interests**: Foundational time-series models, Scalable and robust Time-Series forecasting, Probabilistic Forecasting, Uncertainty Quantification and Epidemic Forecasting

## Education

**Georgia Institute of Technology** — **Atlanta, GA**
*Machine Learning PhD, Computational Science and Engineering* — *Jan 2021- June 2026 (Expected)*
Advisor: B Aditya Prakash

**Indian Institute of Technology Madras** — **Chennai, India**
*Dual Degree (B.Tech + M.tech) in Computer Science and Engineering, GPA: 9.7/10* — *Aug 2015 - Aug 2020*

- Masters advisor: Balaraman Ravindran
- Won awards for best Academic record and best Masters project
- Thesis title: *Learning policies for Social Network Discovery with Reinforcement Learning*

## Research

### Recent Preprints

[P1] *How can time series analysis benefit from multiple modalities? a survey and outlook (2025)*[Paper]
Haoxin Liu, Zhiyuan Zhao, Shangqing Xu, Shiyu Wang, Qingsong Wen, Tom Hartvigsen, Fei Wang, **Harshavardhan Kamarthi**, B. Aditya Prakash

[P2] *PEMs: Pre-trained Epidemic Time-Series Models (2023)*[Paper]
**Harshavardhan Kamarthi**, B. Aditya Prakash

### Conferences

[C1] *Large Pre-trained time series models for cross-domain Time series analysis tasks (2024)*[Paper][Code]
**Harshavardhan Kamarthi**, B. Aditya Prakash;
NeurIPS 2024

[C2] *In-context Pre-trained Time-Series Foundation Models adapt to Unseen Tasks (2025)*[Paper]
Shangqing Xu, **Harshavardhan Kamarthi**, Haoxin Liu, B. Aditya Prakash;
CIKM 2025

[C3] *Time-MMD: A New Multi-Domain Multimodal Dataset for Time Series Analysis (2024)*[Paper][Code]
Haoxin Liu, Shangqing Xu, Zhiyuan Zhao, Lingkai Kong, **Harshavardhan Kamarthi**, Aditya B. Sasanur, Megha Sharma, Jiaming Cui, Qingsong Wen, Chao Zhang, B. Aditya Prakash;
NeurIPS 2024

[C4] *Large Scale Hierarchical Industrial Demand Time-Series Forecasting incorporating Sparsity*[Paper] [Code]
**Harshavardhan Kamarthi**, Aditya B Sasanur, Xinjie Tong, Xingyu Zhou, James Peters, Joseph Czyzyk, B. Aditya Prakash;
KDD Applied Data Science Track 2024

[C5] *Machine learning for data-centric epidemic forecasting* [Paper] [Code]
**Harshavardhan Kamarthi**, Alexander Rodriguez, B. Aditya Prakash;
Nature Machine Intelligence 2024

[C6] *Title evaluation of FluSight influenza forecasting in the 2021âĂŞ22 and 2022âĂŞ23 seasons with a new target laboratory-confirmed influenza hospitalizations*[Paper]
Nature Communications 2024

[C7] *LSTPrompt: Large language models as zero-shot time series forecasters by long-short-term prompting*
Haoxin Liu, Zhiyuan Zhao, Jindong Wang, **Harshavardhan Kamarthi**, B. Aditya Prakash;
ACL Findings 2024

[C8] *Time-Series Forecasting for Out-of-Distribution Generalization Using Invariant Learning*[Paper] [Code]
Haoxin Liu, **Harshavardhan Kamarthi**, Zhiyuan Zhao, Chao Zhang B. Aditya Prakash;

ICML 2024

[C9] *PROFHIT: Probabilistic Robust Forecasting for Hierarchical Time-series* [Paper] [Code]
**Harshavardhan Kamarthi**, Lingkai Kong, Alexander Rodriguez, Chao Zhang, B. Aditya Prakash;
KDD 2023

[C10] *Back2Future: Leveraging Backfill Dynamics for Improving Real-time Predictions in Future* [Paper] [Code]
**Harshavardhan Kamarthi**, Alexander Rodriguez, B Aditya Prakash;
ICLR 2022

[C11] *CAMul: Calibrated and Accurate Multi-view Time-Series Forecasting* [Paper] [Code]
**Harshavardhan Kamarthi**, Lingkai Kong, Alexander Rodriguez, Chao Zhang, B Aditya Prakash;
WWW 2022

[C12] *When in Doubt: Neural Non-Parametric Uncertainty Quantification for Epidemic Forecasting* [Paper] [Code]
**Harshavardhan Kamarthi**, Lingkai Kong, Alexander Rodriguez, Chao Zhang, B Aditya Prakash
Conference on Neural Information Processing Systems (NeurIPS) 2021

[C13] *Influence maximization in unknown social networks: Learning Policies for Effective Graph Sampling* [Paper] [Code]
**Harshavardhan Kamarthi**, Priyesh Vijayan, Bryan Wilder, Balaraman Ravindran, Milind Tambe
International Conference on Autonomous Agents and Multiagent Systems (AAMAS) 2020
**(Nominated for best paper award)**

[C14] *Reinforcement Learning for Unified Allocation and Patrolling in Signaling Games with Uncertainty* [Paper] [Code]
Aravind Venugopal, Elizabeth Bondi, **Harshavardhan Kamarthi**, Keval Dholakia, Balaraman Ravindran, Milind Tambe
International Conference on Autonomous Agents and Multiagent Systems (AAMAS) 2021

[C15] *Integrating Lexical Knowledge in Word Embeddings using Sprinkling and Retrofitting* [Paper]
**Harshavardhan Kamarthi***, Aakash Srinivasan*, Devi Ganesan, Sutanu Chakraborti
International Conference on Natural Language Processing 2020

## Workshops

[W1] *Learning Latent Graph Structures for Accurate and Calibrated Time-series Forecasting* [Paper] [Code]
**Harshavardhan Kamarthi**, Lingkai Kong, Alexander Rodriguez, Chao Zhang, B. Aditya Prakash
UDM Workshop at KDD 2024

[W2] *Selective Intervention Planning using Restless Multi-Armed Bandits to Improve Maternal and Child Health Outcomes*
Siddharth Nishtala, Lovish Madaan, **Harshavardhan Kamarthi**, Anirudh Grama, Divy Thakkar, Dhyanesh Narayanan, Suresh Chaudhary, Neha Madhiwalla, Ramesh Padmanabhan, Aparna Hegde, Pradeep Varakantham, Balaraman Ravindran, Milind Tambe
AAAI 2021 Workshop on AI For Behavior Change

[W3] *Missed calls, Automated Calls and Health Support: Using AI to improve maternal health outcomes by increasing program engagement*
Siddharth Nishtala, **Harshavardhan Kamarthi**, Divy Thakkar, Dhyanesh Narayanan, Anirudh Grama, Aparna Hegde, Ramesh Padmanabhan, Neha Madhiwalla, Suresh Chaudhary, Balaraman Ravindran, Milind Tambe
Harvard CRCS Workshop on AI for Social Good 2020

## Journals

[J1] *Data-Centric Epidemic Forecasting: A Survey* [Paper]
Alexander Rodriguez*, **Harshavardhan Kamarthi***, Pulak Agarwal, Javen Ho, Mira Patel, Suchet Sapre, B. Aditya Prakash
To Appear at Nature Machine Intelligence

[J2] *Evaluation of FluSight influenza forecasting in the 2021âĂŞ22 and 2022âĂŞ23 seasons with a new target laboratory-confirmed influenza hospitalizations* [Paper]
Sarabeth M Mathis, Alexander E. Webber, TomÃąs M. LeÃșn, Erin L. Murray, Monica Sun, Lauren A. White, Logan C. Brooks, Alden Green, Addison J. Hu, Roni Rosenfeld, Dmitry Shemetov, Ryan J. Tibshirani, Daniel J. McDonald, Sasikiran Kandula, Sen Pei, Rami Yaari, Teresa K. Yamana, Jeffrey Shaman, Pulak Agarwal, Srikar Balusu, Gautham Gururajan, Harshavardhan Kamarthi, B. Aditya Prakash, et. al
Nature Communications 2024

### Other Preprints

[OP1] *Efficient language model editing via contextual prompt generator (2022)*[Paper]
**Harshavardhan Kamarthi**, Yen-Chang Hsu, Yilin Shen, Hongxia Jin

## Professional Experiences

**Conviva** **Foster City, CA**
*Research Intern, Host: Vyas Sekar* *May 2024 - Aug 2024*
Research topic: Building a Mixture of Expert models for Root-cause prediction of anomalies in App and Video streaming data

**Conviva** **Foster City, CA**
*Research Intern, Host: Vyas Sekar* *May 2023 - Aug 2023*
Research topic: Real-time Multi-cohort user analytics time-series anomaly detection

**Samsung Research America** **Mountain View, CA**
*Research Intern, Host: Yen-Chang Hsu* *May 2022 - Aug 2022*
Research topic: Semi-Parametric Prompt generation for editing Large Language Models' beliefs

**Microsoft (R&D) Pvt Ltd** **Hyderabad, India**
*Software Development Intern* *May 2018 - July 2018*

**Indian Institute of Science** **Bangalore, India**
*Summer Research Intern* *May 2017 - July 2017*

## Honors and Awards

**Lakshmi Ravi Award**
*Best Masters (Dual Degree) Project at IIT Madras, **1st** out of class of 61* *May 2020*

**Alumni Association Award**
*Best Academic Record in Computer Science at IIT Madras, **1st** out of class of 61* *May 2020*

**Kishore Vaigyanik Protsahan Yojana Fellow**
***National rank 82** out of over 100,000 participants* *2015*

**National Talent Search Scholar**
***0.5%** acceptance out of over 1,000,000* *2012*

## Services

### Talks

**Keynote Talk on Robust TIme-series Forecasting** **KDD 2023**
*Workshop on Uncertainty Reasoning and Quantification in Decision Making*

### Tutorials

**Data-centric Epidemic Forecasting** **KDD 2022**
*Alexander Rodriguez*, **Harshavardhan Kamarthi***, B Aditya Prakash* [Link to slides]

**AI for Data-centric Epidemic Forecasting** **AAAI 2023**
*Alexander Rodriguez*, **Harshavardhan Kamarthi***, B Aditya Prakash* [Link to slides]

### Grant Writing

**NSF Predictive Intelligence for Pandemic Prevention Phase I: Development Grants**
*BEHIVE - BEHavioral Interaction and Viral Evolution for Pandemic Prevention and Prediction*
PI: Dr. B. Aditya Prakash, with collaborators from Georgia Tech, MIT, Univ. of Michigan, Mayo Clinic, and Univ. of Georgia

### Professional

**2021, 2022, 2023, 2025**: NeurIPS *Reviewer*

**2022, 2023, 2025**: ICLR *Reviewer*

**2021, 2023, 2024, 2025**: KDD *Reviewer*

**2023, 2024**: ICML *Reviewer*

**2021, 2022, 2023**: International Workshop on Epidemiology meets Data Mining and Knowledge *Reviewer*

**2021**: National Symposium on Predicting Emergence of Virulent Entities by Novel Technologies (PREVENT) *PR and Tech*

**2020**: IJCAI workshop in AI for Social Good *Reviewer*

**2017**: Software Programming Workshop at IIT Madras *Co-Organizer and Problem Writer*

### Teaching Assistant

**Fall 2022, Fall 2024**: CS8803 Epi: Data Science for Epidemiology (Dr. B Aditya Prakash) *Georgia Tech*

**Fall 2019**: CS6250: Computational Models of Cognition (Dr. Sutanu Chakraborty) *IIT Madras*

**Spring 2020**: CS6700: Reinforcement Learning (Dr. Balaraman Ravindran) *IIT Madras*

## Relevant Courses

Statistical Machine Learning, Probabilistic Graphical Models, Computer Vision, Data science for Epidemiology, Convex Optimization, Principles of Machine Learning, Deep Learning, Reinforcement Learning, Natural Language Processing,

## Technical skills

**Languages**: Python, R, C, C++, Julia, Java

**Tools**: PyTorch, LaTeX, Git, CUDA